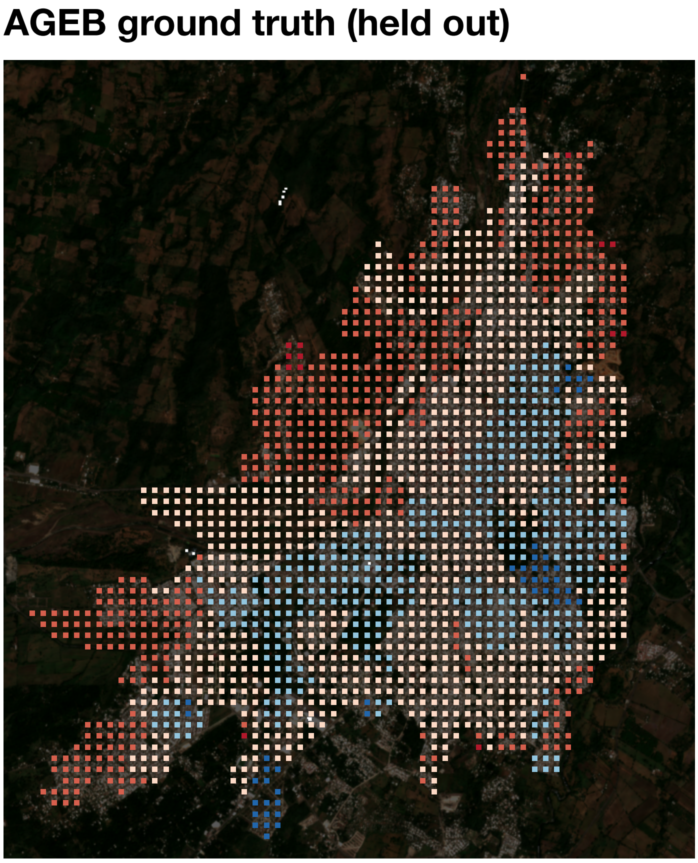
Supplementary Information

Mapping intra-urban social deprivation from Sentinel imagery using only municipal aggregate labels



Supplementary Figure 1: Tract-level ground truth for Tapachula, on the color scale of main-text Fig. 1c. Every token inherits the held-out GRS grade of the AGEB it falls in.

Supplementary Table 1: The supervision-efficiency curve (main-text Fig. 2a). Mean over three seeds; validation cities were used for selection, test cities were scored once with frozen models.

Bags	Validation within-ρ (s.d.)	Test within-ρ
50	0.150 (0.021)	0.156
100	0.141 (0.021)	0.158
200	0.151 (0.022)	0.161
400	0.182 (0.020)	0.192
771	0.227 (0.010)	0.185 (retrained); 0.196 (saved weights)

Supplementary Table 2: Agreement by tract size (main-text Limits). Within-municipality Spearman over municipalities with at least ten tracts in the bin.

Tokens per tract	Tracts	Median population	Municipalities	Within-ρ
1–3	567	211	15	0.257
4–9	849	1,948	16	0.338
10–19	999	2,644	20	0.352
20–49	464	3,231	14	0.432
$\geq$50	79	3,561	2	0.127

Supplementary Table 3: Municipalities excluded from the expansion for insufficient optical depth or persistent acquisition errors; the guard requires eight observations per pixel and excludes shallow ground instead of compositing it.

Municipality	Reason
Amozoc; Comalcalco; Hidalgo del Parral; Lagos de Moreno; Nuevo Casas Grandes	composite below the eight-observation depth floor (optical arm)
La Independencia	inconsistent tract geometry
El Marqués	repeated acquisition timeouts

Supplementary Table 4: Free global products under the map's own protocol. Within-ρ is the mean over municipalities of the within-municipality Spearman correlation against the held-out tract grade, on the identical universe; Δ is the paired difference with a city-clustered bootstrap interval; AUROC is pooled detection of high and very-high grade tracts across the split. Orientation was fixed before measurement: deprivation is the negative of every product. The nighttime-light row rests on interior-point sampling for 63% of tracts, which are smaller than one of its 30 arcsec cells.

Product	Split	Map	Product	Δ [95% CI]	AUROC map / product
GHSL built-up surface	val	0.354	0.202	+0.152 [−0.103, +0.303]	0.828 / 0.866
	test	0.232	0.174	+0.059 [−0.098, +0.160]	0.829 / 0.908
GHSL building height	val	0.354	0.226	+0.128 [−0.096, +0.291]	0.828 / 0.859
	test	0.232	0.119	+0.113 [−0.062, +0.233]	0.829 / 0.886
GHSL population	val	0.354	0.065	+0.289 [−0.001, +0.497]	0.828 / 0.872
	test	0.232	0.045	+0.188 [+0.051, +0.289]	0.829 / 0.855
Nighttime lights	val	0.364	0.327	+0.037 [−0.109, +0.149]	0.828 / 0.840
	test	0.210	0.249	−0.039 [−0.245, +0.104]	0.829 / 0.831

Supplementary Table 5: Conformal grade intervals at 90% nominal coverage, calibrated on the validation cities and evaluated on the test cities. The marginal rule assumes all tracts exchangeable; the clustered rule takes each calibration city's quantile and then the 90th percentile across cities.

Rule	Half-width (grades)	Coverage	Worst city	Worst city coverage	Cities below
Marginal	1.29	0.869	La Paz	0.768	10 of 14
Clustered by city	1.68	0.953	Madera	0.870	1 of 14

Supplementary Table 6: The fourteen test cities ranked by the within-municipality correlation their map achieves, worst first. Tract-mean scores. Spread is the median across tracts of the standard deviation over three seeds, which does not separate the failures from the successes.

City	Municipalities	Tracts	Within-ρ	Median seed spread
Guerrero	1	21	-0.510	0.101
Ciudad del Carmen	1	54	-0.267	0.069
Escárcega	1	24	-0.001	0.093
Madera	1	46	0.052	0.133
Querétaro	3	514	0.091	0.109
Zapopan	2	335	0.105	0.104
García	3	324	0.257	0.088
Nogales	1	171	0.261	0.093
Victoria	1	165	0.373	0.102
Hidalgo	1	71	0.425	0.098
Irapuato	1	214	0.465	0.121
Ixtapaluca	4	463	0.471	0.097
La Paz	1	276	0.534	0.095
Iguala de la Independencia	1	133	0.639	0.125